Do Now · retrieval

Programming constructs and scope

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the approximate growth of a loop executed n times?

Q2

Why must binary search update low=mid+1 rather than low=mid in the greater-than case?

Q3

What is true AND false?

Q4

What does int a=2; a=a+3; Console.WriteLine(a); output?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Three control structures — Sequence runs in order; selection chooses a branch; iteration repeats instructions.

Core idea 2

Where is a variable visible? — A local declaration belongs to its block or method; a class-level field can be accessed by relevant class methods.

Core idea 3

Shadowing — A local variable with the same name as a field can hide that field inside the method; trace precisely which one is printed.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

sequence — Operations execute in order.

selection — Execution chooses a branch according to a condition.

iteration — A block repeats while a condition permits.

scope — Where a name can be accessed.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

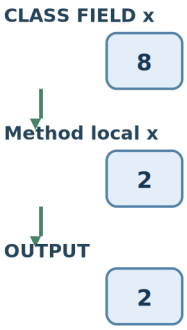
A class field x is 8. A method declares int x=2 and prints x. What appears?

Follow the worked method

1. Predict first — Which x is in scope at the print statement?

2. Model solution — The method-local x is 2 and shadows the field. The program prints 2.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
Which variable is modified?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Class field level = 9; method declares int level = 4 and prints level. What appears?

Question 2

Class field attempts = 2; method executes attempts++; then prints attempts. What appears?

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Describe one risk of unnecessary global/class-level mutable variables.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A local variable always changes a same-named class field.

Use key vocabulary from memory

1. Explain sequence in your own words.

2. Explain selection in your own words.

3. Explain iteration in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.